

Finite Element Approximation of a Stochastic Advection–Diffusion Equation with Additive Noise

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1 Weak Formulation of the Stochastic Advection–Diffusion Problem

1.1 Strong Formulation

Let $\Omega \subset \mathbb{R}^2$ be a bounded polygonal domain with boundary $\partial\Omega$, and let $T > 0$ be a final time.

We consider the stochastic advection–diffusion equation

$$du = (-\mathbf{b} \cdot \nabla u + \nabla \cdot (\kappa \nabla u) + f) dt + \sigma, dW_t \quad \text{in } \Omega \times (0, T), \quad (1)$$

subject to the homogeneous Dirichlet boundary condition

$$u = 0 \quad \text{on } \partial\Omega \times (0, T), \quad (2)$$

and the initial condition

$$u(x, 0) = u_0(x) \quad \text{in } \Omega. \quad (3)$$

Here,

- $u = u(x, t)$ denotes the unknown state variable,
- $\mathbf{b} = (b_1, b_2)^T$ is a prescribed velocity field,
- $\kappa > 0$ is the diffusion coefficient,
- f is a source term,
- $\sigma > 0$ is the noise intensity,
- $W_t * t \geq 0$ is a standard Wiener process defined on a filtered probability space $(\Omega_P, \mathcal{F}, \mathcal{F}_t * t \geq 0, \mathbb{P})$.

1.2 Assumptions

Throughout this work, we assume that

1. $\kappa > 0$;
2. $\mathbf{b} \in [L^\infty(\Omega)]^2$;
3. $f \in L^2(0, T; L^2(\Omega))$;
4. $u_0 \in L^2(\Omega)$.

Moreover, for simplicity, we assume that

$$\nabla \cdot \mathbf{b} = 0 \quad \text{in } \Omega.$$

1.3 Functional Setting

Let

$$H = L^2(\Omega),$$

equipped with the inner product

$$(u, v)_H = \int_{\Omega} uv, dx,$$

and norm

$$|u|_H = (u, u)_H^{1/2}.$$

We define

$$V = H_0^1(\Omega),$$

endowed with the norm

$$|v|_V = |\nabla v|_{L^2(\Omega)}.$$

The Gelfand triple

$$V \subset H \subset V'$$

will be used throughout the analysis.

1.4 Weak Formulation

Let $v \in V$ be a test function.

Multiplying equation (1) by v and integrating over Ω , we obtain

$$(du, v) = (-\mathbf{b} \cdot \nabla u, v), dt + (\nabla \cdot (\kappa \nabla u), v), dt + (f, v), dt + \sigma(v), dW_t.$$

Using Green's formula and the homogeneous boundary condition, we have

$$(\nabla \cdot (\kappa \nabla u), v) = -\kappa(\nabla u, \nabla v).$$

Therefore, the weak formulation becomes

$$(du, v) + a(u, v), dt = (f, v), dt + \sigma(v), dW_t,$$

for all $v \in V$, where the bilinear form

$$a : V \times V \rightarrow \mathbb{R}$$

is defined by

$$a(u, v) = \kappa(\nabla u, \nabla v) + (\mathbf{b} \cdot \nabla u, v).$$

1.5 Definition of Weak Solution

A stochastic process

$$u : [0, T] \times \Omega_P \rightarrow V$$

is called a weak solution of problem (1) if

$$u \in L^2(\Omega_P; L^2(0, T; V)),$$

is adapted to the filtration $\mathcal{F}_{t \geq 0}$, satisfies the initial condition

$$u(0) = u_0,$$

and

$$(u(t), v) + \int_0^t a(u(s), v), ds = (u_0, v) + \int_0^t (f(s), v), ds + \sigma \int_0^t (v), dW_s,$$

for every test function $v \in V$ and almost surely.

2 Energy Estimate and Stability

In this section, we derive an a priori energy estimate for the weak solution of the stochastic advection–diffusion equation.

2.1 Itô Formula

Let

$$\Phi(u) = \|u\|_{L^2(\Omega)}^2.$$

Applying Itô's formula to the stochastic process

$$du = (Au + f) dt + \sigma dW_t,$$

we obtain

$$d\|u\|^2 = 2(Au, u) dt + 2(f, u) dt + \sigma^2 dt + 2\sigma(u, dW_t).$$

2.2 Evaluation of the Deterministic Terms

Recall that

$$Au = -\mathbf{b} \cdot \nabla u + \nabla \cdot (\kappa \nabla u).$$

Hence

$$(Au, u) = -(\mathbf{b} \cdot \nabla u, u) + (\nabla \cdot (\kappa \nabla u), u).$$

Assuming that

$$\nabla \cdot \mathbf{b} = 0,$$

and using the homogeneous boundary condition, we obtain

$$(\mathbf{b} \cdot \nabla u, u) = 0.$$

Furthermore,

$$(\nabla \cdot (\kappa \nabla u), u) = -\kappa \|\nabla u\|^2.$$

Consequently,

$$(Au, u) = -\kappa \|\nabla u\|^2.$$

2.3 Energy Equality

Substituting into the Itô formula yields

$$d\|u\|^2 + 2\kappa \|\nabla u\|^2 dt = 2(f, u) dt + \sigma^2 dt + 2\sigma(u, dW_t).$$

Using Young's inequality,

$$2(f, u) \leq \|f\|^2 + \|u\|^2.$$

Therefore,

$$d\|u\|^2 + 2\kappa \|\nabla u\|^2 dt \leq (\|u\|^2 + \|f\|^2 + \sigma^2) dt + 2\sigma(u, dW_t).$$

2.4 Expectation

Integrating from 0 to t and taking expectations gives

$$\begin{aligned} & \mathbb{E}[\|u(t)\|^2] + 2\kappa \mathbb{E} \left[\int_0^t \|\nabla u(s)\|^2 ds \right] \\ & \leq \|u_0\|^2 + \int_0^t \mathbb{E}[\|u(s)\|^2] ds + \int_0^t \|f(s)\|^2 ds + \sigma^2 t. \end{aligned}$$

Since

$$\mathbb{E} \left[\int_0^t (u(s), dW_s) \right] = 0,$$

the stochastic integral disappears after taking expectations.

2.5 A Priori Estimate

Applying Grönwall's lemma, we obtain

$$\mathbb{E}[\|u(t)\|^2] \leq C \left(\|u_0\|^2 + \int_0^T \|f(s)\|^2 ds + \sigma^2 T \right),$$

where $C > 0$ depends only on T .

Theorem 2.1. *Let u be a weak solution of the stochastic advection–diffusion problem. Then*

$$\sup_{0 \leq t \leq T} \mathbb{E}[\|u(t)\|^2] + \kappa \mathbb{E} \left[\int_0^T \|\nabla u(t)\|^2 dt \right] \leq C,$$

where

$$C = C(T, \kappa, \|u_0\|^2, \|f\|_{L^2(0,T;L^2(\Omega))}).$$

This estimate provides the fundamental stability property required for the existence analysis and the finite element approximation developed in the following sections.

3 Existence and Uniqueness of Weak Solutions

In this section, we establish the well-posedness of the stochastic advection–diffusion problem introduced in the previous section.

3.1 Continuity of the Bilinear Form

Recall that the bilinear form is defined by

$$a(u, v) = \kappa(\nabla u, \nabla v) + (\mathbf{b} \cdot \nabla u, v),$$

for all $u, v \in V$.

Lemma 3.1. *The bilinear form $a(\cdot, \cdot)$ is continuous on $V \times V$. More precisely, there exists a constant $C > 0$ such that*

$$|a(u, v)| \leq C \|u\|_V \|v\|_V$$

for all $u, v \in V$.

Proof. Using the Cauchy–Schwarz inequality,

$$|\kappa(\nabla u, \nabla v)| \leq \kappa \|\nabla u\| \|\nabla v\|.$$

Moreover,

$$|(\mathbf{b} \cdot \nabla u, v)| \leq \|\mathbf{b}\|_{L^\infty} \|\nabla u\| \|v\|.$$

Applying the Poincaré inequality,

$$\|v\| \leq C_P \|\nabla v\|,$$

we obtain

$$|a(u, v)| \leq C \|u\|_V \|v\|_V.$$

Hence the result follows. $\square$

3.2 Coercivity

Lemma 3.2. *Assume that*

$$\nabla \cdot \mathbf{b} = 0.$$

Then the bilinear form $a(\cdot, \cdot)$ is coercive on V :

$$a(v, v) \geq \kappa \|\nabla v\|^2$$

for all $v \in V$.

Proof. We have

$$a(v, v) = \kappa \|\nabla v\|^2 + (\mathbf{b} \cdot \nabla v, v).$$

Since $\nabla \cdot \mathbf{b} = 0$ and $v = 0$ on $\partial\Omega$,

$$(\mathbf{b} \cdot \nabla v, v) = 0.$$

Therefore,

$$a(v, v) = \kappa \|\nabla v\|^2.$$

Since $\kappa > 0$, the result follows. $\square$

3.3 Existence of Weak Solutions

We are now in a position to establish the existence of a weak solution.

Theorem 3.3. *Assume that*

$$u_0 \in L^2(\Omega),$$

$$f \in L^2(0, T; L^2(\Omega)),$$

and

$$\mathbf{b} \in [L^\infty(\Omega)]^2.$$

Then there exists a weak solution

$$u \in L^2(\Omega_P; L^2(0, T; V))$$

to the stochastic advection–diffusion problem.

Sketch of proof. The proof is based on the Galerkin approximation method.

Let

$$V_m = \text{span}\{\varphi_1, \dots, \varphi_m\}.$$

be a finite-dimensional subspace of V .

We seek an approximation

$$u_m(t) = \sum_{i=1}^m c_i(t) \varphi_i.$$

The coefficients satisfy a finite-dimensional stochastic system.

Using the a priori estimate established in Section 3, one obtains uniform bounds with respect to m .

Compactness arguments and weak convergence allow passage to the limit.

Hence a weak solution exists. □

3.4 Uniqueness

Theorem 3.4. *The weak solution of the stochastic advection–diffusion problem is unique.*

Proof. Let u_1 and u_2 be two weak solutions.

Define

$$w = u_1 - u_2.$$

Subtracting the two equations yields

$$dw = Aw \, dt.$$

Testing with w gives

$$\frac{1}{2} \frac{d}{dt} \|w\|^2 + a(w, w) = 0.$$

Using coercivity,

$$\frac{1}{2} \frac{d}{dt} \|w\|^2 + \kappa \|\nabla w\|^2 = 0.$$

Since

$$w(0) = 0,$$

we obtain

$$\|w(t)\|^2 = 0$$

for all $t \in [0, T]$.

Therefore

$$u_1 = u_2.$$

Hence the solution is unique. □

4 Finite Element Approximation and Convergence Analysis

In this section, we introduce the finite element approximation of the stochastic advection–diffusion problem and derive a convergence estimate.

4.1 Finite Element Space

Let $\mathcal{T}_h$ be a regular triangulation of the domain Ω with mesh size h .

We define the finite element space

$$V_h = \{v_h \in H_0^1(\Omega) : v_h|_K \in \mathbb{P}_1(K), \quad \forall K \in \mathcal{T}_h\}.$$

Throughout this work, we assume that the mesh family is shape-regular and quasi-uniform.

4.2 Ritz Projection

Let

$$R_h : V \rightarrow V_h$$

denote the Ritz projection defined by

$$a(R_h u, v_h) = a(u, v_h), \quad \forall v_h \in V_h.$$

The following approximation properties are classical.

Lemma 4.1. *Assume that*

$$u \in H^2(\Omega) \cap H_0^1(\Omega).$$

Then there exists a constant $C > 0$ independent of h such that

$$\|u - R_h u\|_{L^2(\Omega)} \leq Ch^2 \|u\|_{H^2(\Omega)},$$

and

$$\|\nabla(u - R_h u)\|_{L^2(\Omega)} \leq Ch \|u\|_{H^2(\Omega)}.$$

4.3 Semi-Discrete Formulation

The semi-discrete finite element approximation seeks

$$u_h(t) \in V_h$$

such that

$$(du_h, v_h) + a(u_h, v_h) dt = (f, v_h) dt + \sigma(v_h) dW_t,$$

for all

$$v_h \in V_h.$$

Let

$$\{\varphi_i\}_{i=1}^{N_h}$$

be a basis of V_h . Writing

$$u_h = \sum_{j=1}^{N_h} U_j(t) \varphi_j,$$

we obtain the matrix system

$$M dU + A U dt = F dt + \sigma G dW_t,$$

where

$$M_{ij} = (\varphi_j, \varphi_i),$$

is the mass matrix and

$$A_{ij} = a(\varphi_j, \varphi_i),$$

is the stiffness-advection matrix.

4.4 Fully Discrete Scheme

Let

$$t_n = n\Delta t, \quad n = 0, \dots, N.$$

Using the implicit Euler scheme, we obtain

$$MU^{n+1} + \Delta t AU^{n+1} = MU^n + \Delta t F^n + \sigma G \Delta W_n,$$

where

$$\Delta W_n = W_{t_{n+1}} - W_{t_n}.$$

The Wiener increments satisfy

$$\Delta W_n \sim \mathcal{N}(0, \Delta t).$$

4.5 Discrete Stability

Theorem 4.2. *Assume that the hypotheses of Sections 2 and 3 hold.*

Then the fully discrete solution satisfies

$$\max_{0 \leq n \leq N} \mathbb{E} [\|u_h^n\|^2] + \Delta t \sum_{n=0}^N \mathbb{E} [\|\nabla u_h^n\|^2] \leq C,$$

where the constant C is independent of h and Δt .

Sketch of proof. Choosing

$$v_h = u_h^{n+1}$$

in the discrete formulation and using

$$(a - b)a = \frac{1}{2} (a^2 - b^2 + (a - b)^2),$$

one derives a discrete energy inequality.

Taking expectations and applying a discrete Grönwall lemma yields the result.

□

4.6 Error Decomposition

Let

$$e^n = u(t_n) - u_h^n.$$

We decompose the error as

$$e^n = \underbrace{u(t_n) - R_h u(t_n)}_{\eta^n} + \underbrace{R_h u(t_n) - u_h^n}_{\theta^n}.$$

The term η^n represents the interpolation error, while θ^n is the discrete error. Using the approximation properties of the Ritz projection, we obtain

$$\|\eta^n\| \leq Ch^2.$$

4.7 Convergence Estimate

We now state the main convergence result.

Theorem 4.3. *Assume that the exact solution satisfies*

$$u \in L^2(\Omega_P; H^2(\Omega)),$$

and possesses sufficient temporal regularity.

Then there exists a constant $C > 0$, independent of h and Δt , such that

$$\max_{0 \leq n \leq N} \mathbb{E} [\|u(t_n) - u_h^n\|^2] \leq C (h^2 + \Delta t).$$

Sketch of proof. Subtracting the continuous and discrete formulations leads to an equation for the discrete error θ^n .

Using the coercivity of the bilinear form, the stability estimate, and the approximation properties of the Ritz projection, we obtain

$$\mathbb{E} [\|\theta^n\|^2] \leq C (h^2 + \Delta t).$$

Combining this estimate with the interpolation error yields the desired result. □

The above theorem shows that the proposed finite element approximation converges strongly in mean square as

$$h, \Delta t \rightarrow 0.$$